University of Central Punjab

Faculty of Information Technology and Computer Science

Course Title: Human Computer Interaction
Course Code: SESE 2123

Assignment 2

Date: 08-11-2025

Total Marks 50 (Assignment) +10 (Evaluation)=60

Name	Registration Number	Section
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CLO #	Course Learning Outcome (CLO)	Taxonomy Level	Mapping to PLO
CLO 2	Apply user centric methods to understand and analyze population dynamics, their wants, needs and irritations.	C3	PLO 3

Instructions:

1. Attempt all questions.
2. Group Member Allowed (Maximum 2)
3. Assignment should be handwritten and submitted on A4 sheets or Assignment sheets only. Violation will result to deduction of 1 mark from the scored marks.
4. Each student will have attached this front page with his/her assignment. Violation will result to deduction of 1 mark from the scored marks.
5. Due Date for Assignment is November 14, 2025 10:00 PM (Portal).
6. Late submission will result in 10% deduction in marks.
7. No request for late submissions will be considered after two working days of the deadline.
8. No request to review assignment will be considered after 3 working days of the review in class.

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Q#1 Brainstorming

a) Brainstorming is a collaborative, open ended ideation technique that encourages the free flow of ideas without immediate judgment or criticism, fostering creativity and diversity in thinking. For the AI fitness app design team, it would serve multiple key purposes:

- **Uncovering hidden user needs:** By discussing pain points like lack of personalization, motivational fatigue, and disconnected lifestyle factors team members can rapidly generate ideas that directly address these, such as AI-driven adaptive routines
- **Encouraging cross-disciplinary input:** Software engineering students could combine technical expertise (e.g., AI algorithms, wearable integration) with user empathy, leading to innovate features like gamified process tracking or voice-based encouragement that mimics human coaching.
- **Boosting team morale and ownership:** Inclusive sessions make all members contribute, leading to higher commitment during design and development phases.
- **Building on ideas Iteratively:** The no bad ideas rule allows initial concepts (e.g., basic calorie tracking) to evolve into advanced ones (e.g., predictive AI that Anticipates motivation dips based on sleep data), sparking breakthroughs in personalization and usabil

Prioritizating feasibility and differentiation: Post-session evalu would help filter ideas against constraints like privacy and accessibility, ensuring solutions are inno yet practical, reducing the risk of generic feature seen in existing apps.

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b) Five Creative ideas

1. **AI 'Mood Mentor' Avatar:** A customizable, animated AI coach that uses wearable data (heart rate, sleep patterns) and user mood logs to deliver empathetic, human-like encouragment via voice or text, e.g., "I see you're stressed today - let's do 10-min gentle yoga to recharge!" This combats motivationals fatigue with tailored, conversational reminders.

2. **Holistic "Life Web" Dashboard:** An interactive visual app connecting diet, exercise, sleep, and stress data in real-time, with AI explanations like "Your poor sleep last night reduced workout stamina by 20% - try this quick meal tweak." Users drag element to simulate changes, making overwhelming datd intuitive and educational.

3. **Progressive 'Skill tree' Gamification:** Modeled after video games, users unlock personlized "branches" (e.g., stamina, flexibility) based on physical abilities and medical conditions, earning rewards like custom badges or virtual trainer upgrades. AI adjusts difficulty dynamically to prevent dropout, differing from rigid plans.

4. **Privacy-First 'Data Vault':** Users store sensitive data in an encypted personal vault, granting AI temporary access only for specific queries (e.g., "Analyze my week"). The app provides transparent logs of data usage and anonymized community benchmarks, building trust while enabling secure personalization.

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5- **Beginner - Friendly with AR Overlays:** Using phone camera for augmented (AR), the app overlays simple exercise demos on the user's environment, with voice-guided corrections for form. For middle-aged or confused users, it starts with ultra-simple interfaces (large buttons, one-tap logging) and gradually introduces features, reducing accessibility barriers.

Q#2 Target Market:

The target Market for the AI fitness app primarily consists of young professionals (aged 22-35) and university students (aged 18-25) in urban or semi-urban areas, who are tech-savvy but time-constrained due to demanding work/study schedules. This group is experiencing rising issues like obesity, anxiety, and poor sleep from sedentary lifestyles, irregular meals, and high stress. They often start gym memberships or generic apps but abandon them due to lack of motivation, personalization, and integrated support for diet/sleep/stress. Secondary segments include middle-aged beginners (aged 35-50) seeking accessible entry into fitness without overwhelmin interfaces, prioritizing simplicity, privacy, and guidance for medical conditions.

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Q:3. User classes:-

a) Predict User classes:

1. University Students

It is primary user

Young adults struggling with irregular routines, stress and motivation for consistent exercise.

2. Working Professionals

primary user

Office workers with busy lifestyles who want to manage weight, stress and sleep.

3. Middle-aged Fitness Beginners

Secondary user

Older users starting fitness journeys who may face usability or accessibility challenges.

b) 1. University Students:

Main Fitness Goal:

Improve stamina, build a sustainable, healthy routine that mitigates stress, improves sleep and prevents lifestyle-related health issues rather than just intense bodybuilding

Tech Familiarity: High. They are digital natives, comfortable with smartphones, wearable sensors and app based solutions. They expect seamless integration and a modern user experience.

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Support Expected from App:-

Personalized training plans, regular reminders, gamified motivation and AI-based feedback.

2. Working Professionals:

Main Fitness Goal:- Maintain a healthy routine, manage weight and reduce anxiety

Tech Familiarity: Moderate to High. use apps regularly but have limited time to explore features.

Support Expected from App: Quick progress summaries, easy progress tracking and integration of stress and sleep management features.

3. Middle-aged Fitness Beginners:-

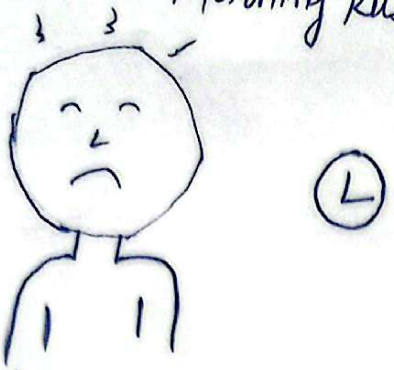
Main Fitness goal: Achieve basic fitness journey and start a journey safely and simply without intimidation. Their goal is often general health improvement, weight management and building basic fitness literacy.

Tech Familiarity: Low to Moderate They may find complex apps overwhelming. They need large buttons, simple navigation, clear instructions

Supported Expected: Simple user interface, voice guidance, clear step-by-step instructions and responsive customer support

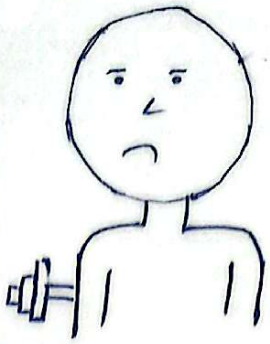
Q:4. Storyboard

Morning Rush



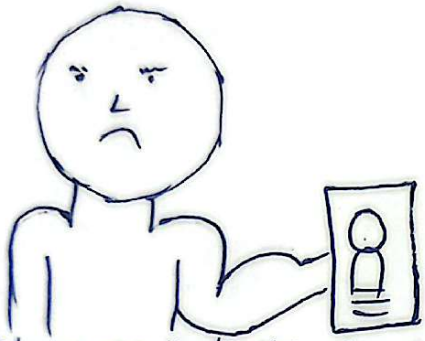
A student wakes up late and skip breakfast before classes

Gym Attempt



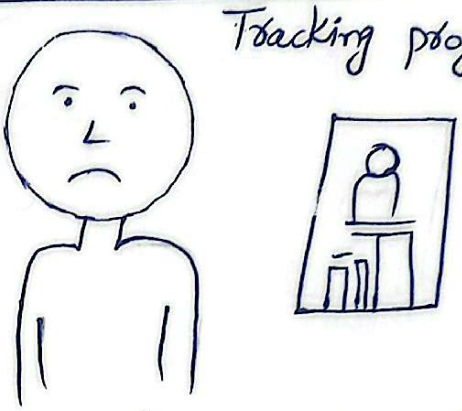
Joins a gym but doesn't know which exercise suit them

Using Fitness App



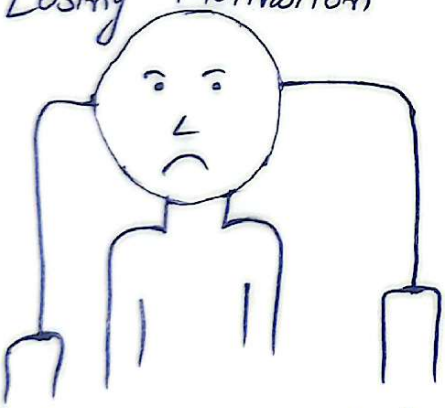
Tries a fitness app that gives generic workout plans

Tracking progress



Sees confusing graphs and Claire numbers

Losing Motivation



Stops exercising regularly after 2 weeks

Health Concern



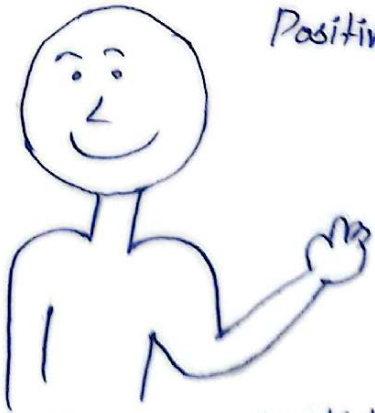
Notices increased stress and poor sleep

Discovery of New AI App



Finds an AI-powered fitness app that gives

Positive change



The AI app tracts diet, sleep and workouts

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Q#5. Need Finding:-

a) Technique to Gather Requirements:-

To gather requirements for the innovative ideas generated in Q1 (like the AI "Mood Mentor", "Holistic Life Web, Skill Tree Gamification, Privacy-First Vault and AR Onboarding Wizard), the most effective Need Finding Technique would be a Combination of Contextual Inquiry and Semi-Structured User Interviews.

1. Contextual Inquiry:

What it is: Observing users in their real environments to understand their daily fitness behaviors, habits and frustrations.

Why use it here: It helps it design team see what users actually do rather than what they say they do.

Example: Watching how users interact with current fitness apps or wearable devices to identify confusion points (e.g., too many menus, unclear charts).

Outcome: Reveals practical insights that guide the UI/UX design of ideas like the AR Onboarding Wizard or Life Web dashboard

2. Semi-Structured Interviews:

What it is: Guided conversations where key questions are prepared, but where users are free to explain their thoughts and feelings in depth.

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Why use it: Guided conversations where key questions uncovers emotional and motivational needs. why users lose interest, what encouragement feels personal what privacy fears they have.

Example: Talking with users about how they would feel if an AI avatar("Mood Mentor") spoke to them daily or analyzed their sleep data.

Outcome: Helps the team refine tones, privacy settings and personalization features.

This technique is best because the ideas in Q1 are: user-centered and emotion-driven, direct-observation plus conversation gives the design team

- Behavioral insights (what users do)
- Emotional insights (why they do it)

b) How Interviews Help:-

Interviews give real insights into users lives, routines, and frustration. They help identify hidden needs, such as motivation loss, confusion about data or privacy fears. Designers can use this information to make the app more personalized, engaging and trustworthy.

Five Interview Questions:

1. What challenges do you face when trying to stay consistent with your fitness routine?

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2. What do you find confusing or frustrating in current fitness apps?

3. How would you feel about an AI coach that gives voice or text encouragement daily?

4. Would visual links between your sleep, diet and workouts help you understand your health better?

5. What level of controls or privacy would make you comfortable sharing your health data?

Q:6. Class Profile (University Students)

Primary User class

Age: 20-25 years

Gender: Male or Female

Job Titles: University Student

Experience Level: Beginners in fitness

Work Hours: 6-8 hours daily

Education: Undergraduate Level

Location: Urban city

Income: Low dependent on parents

Technology: High internet, smartphones, watches

Disabilities: None

Family: Mostly single

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Secondary User (Middle-aged Fitness Beginner):

Age: 35-50 years
Gender: Male or Female
Job Titles: Office workers, Managers, Entrepreneurs
Experience Level: Beginner
Work Hours: 8-10 hours/day
Education: College graduates
Location: Urban or suburban homes
Income: Medium to High
Technology: High uses smartphone, less familiar with fitness app
Disabilities: Some may have joint pain
Family: Married (children), may be single too.

Q#7 Persona



a) Persona for one potential user

1- Header:

- Name: Aisha Rahman
- Summary Quote: "I want to feel energized without the guilt of quitting again."

2- Demographics:

- Personal Background: 24-year-old female, single, lives in a shared apartment in Lahore, Pakistan. South Asian ethnicity, moderate income (PKR 50,000/monthly) from part-time job. Height: 5'4", weight: 68kg (slightly overweight, wants to build stamina for daily commute
- Professional Background: Final-year computer science university student, juggling assignments, internships and freelance coding gigs. Works 10-12 hours/day on laptop, leading to poor posture and irregular sleep (averages 5-6 hours/night).
- User Environment: Uses an Android smartphone and basic fitness tracker (e.g., Mi Band). Studies/exercises at home or campus gym; eats quick hostel meals or delivery food. High-screen time environment with notifications from social media and study Apps; urban pollution adds to fatigue

• **Psychographics:** Aisha is ambitious yet self-critical, launching fitness plans with high energy only to abandon them in 2-3 weeks when progress feels invisible or data overwhelms her; she craves efficient, hyper-personalized guidance and recoils from generic templates. Deeply privacy-conscious, she fears data leaks and needs transparent security to trust the app. She responds best to warm, friend-like encouragement rather than cold metrics, remains largely sedentary with inconsistent calorie logging, and skips meals under stress - yet stays open to AI if it feels intuitive, non-judgmental, and seamlessly weaves diet, sleep into clear, motivating insights.

• **End-Goals:** Aisha aims to establish daily 20-30 min workouts tailored to her energy and schedule, improve sleep to 7+ hours nightly, and lower anxiety via stress-aware routines. She seeks sustainable 10 kg weight loss over six months, enhanced stamina for academic focus, and ongoing motivation through friendly AI coaching.